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# User manual

(Onvif Server)

*HAPPYTIME SOFTWARE CO., LIMITED*

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## Declaration

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Since the product version upgrade or other reasons, this manual will subsequently be updated. Unless otherwise agreed, this manual only as a guide, this manual all statements, information, recommendations do not constitute any express or implied warranties.

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[www.happytimesoft.com](http://www.happytimesoft.com)

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## Chapter 1 Files Description

**Windows** version contains the following files:

| File name             | Description                                                                                                                                                                               |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| onvif.cfg             | The onvif server default configuration file.<br>(A runtime configuration file onvifrun.cfg will be generated when the onvif server stop)                                                  |
| happytime-rtsp-server | Happytime rtsp server. It can stream several kinds of media file.<br>(The rtsp server is a demo version, only for testing rtsp streams, the release version does not include rtsp server) |
| OnvifServer.exe       | onvif server executable file                                                                                                                                                              |
| runme.bat             | A batch file, run rtsp server and onvif server                                                                                                                                            |
| snapshot.jpg          | The default snapshot file, for onvif snapshot interface                                                                                                                                   |
| User manual.pdf       | This manual                                                                                                                                                                               |
| libcrypto-3.dll       | Openssl dynamic library                                                                                                                                                                   |
| libssl-3.dll          | Openssl dynamic library                                                                                                                                                                   |
| cert.pem              | Openssl certificate<br>(For testing only, please apply for an official SSL certificate for use)                                                                                           |
| key.pem               | Openssl private key<br>(For testing only, please apply for an official SSL certificate for use)                                                                                           |

**Linux** version contains the following files:

(Tested on centos 7 and ubuntu 18.04 LTS)

| File name             | Description                                                                                                                                                                               |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| onvif.cfg             | The onvif server default configuration file.<br>(A runtime configuration file onvifrun.cfg will be generated when the onvif server stop)                                                  |
| happytime-rtsp-server | Happytime rtsp server. It can stream several kinds of media file.<br>(The rtsp server is a demo version, only for testing rtsp streams, the release version does not include rtsp server) |
| onvifserver           | onvif server executable file                                                                                                                                                              |
| runme.sh              | A script file, run rtsp server and onvif server                                                                                                                                           |
| snapshot.jpg          | The default snapshot file, for onvif snapshot interface                                                                                                                                   |
| User manual.pdf       | This manual                                                                                                                                                                               |
| libcrypto.so.3        | Openssl dynamic library                                                                                                                                                                   |
| libssl.so.3           | Openssl dynamic library                                                                                                                                                                   |

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|          |                                                                                                 |
|----------|-------------------------------------------------------------------------------------------------|
| cert.pem | Openssl certificate<br>(For testing only, please apply for an official SSL certificate for use) |
| key.pem  | Openssl private key<br>(For testing only, please apply for an official SSL certificate for use) |

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## Chapter 2 Configuration

Onvif server uses default configuration file onvif.cfg.

### 2.1 Configuration Templates

```
<?xml version="1.0" encoding="utf-8"?>
<config>
    <server_ip></server_ip>
    <http_enable>1</http_enable>
    <http_port>8000</http_port>
    <https_enable>1</https_enable>
    <https_port>8443</https_port>
    <cert_file>cert.pem</cert_file>
    <key_file>key.pem</key_file>
    <http_max_users>16</http_max_users>
    <ipv6_enable>1</ipv6_enable>
    <need_auth>0</need_auth>
    <log_enable>1</log_enable>
    <log_level>1</log_level>

    <information>
        <Manufacturer>Happytimesoft</Manufacturer>
        <Model>IPCamera</Model>
        <FirmwareVersion>2.4</FirmwareVersion>
        <SerialNumber>123456</SerialNumber>
        <HardwareId>1.0</HardwareId>
    </information>

    <user>
        <username>admin</username>
        <password>admin</password>
        <userlevel>Administrator</userlevel>
    </user>
```

---

```
<profile>
  <video_source>
    <width>1280</width>
    <height>720</height>
  </video_source>
  <video_encoder>
    <width>1280</width>
    <height>720</height>
    <quality>4</quality>
    <session_timeout>10</session_timeout>
    <framerate>25</framerate>
    <encoding_interval>1</encoding_interval>
    <bitrate_limit>2048</bitrate_limit>
    <encoding>H264</encoding>
    <h264>
      <gov_length>25</gov_length>
      <h264_profile>Main</h264_profile>
    </h264>
  </video_encoder>

  <audio_source></audio_source>
  <audio_encoder>
    <session_timeout>10</session_timeout>
    <sample_rate>8</sample_rate>
    <bitrate>64</bitrate>
    <encoding>G711</encoding>
  </audio_encoder>
  <stream_uri append_params="0"></stream_uri>
</profile>

<scope>onvif://www.onvif.org/location/country/china</scope>
<scope>onvif://www.onvif.org/name/IP-Camera</scope>
<scope>onvif://www.onvif.org/hardware/HI3518C</scope>
```

---

```
<event>
    <renew_interval>60</renew_interval>
    <simulate_enable>1</simulate_enable>
</event>
</config>
```

## 2.2 Configuring Node Description

### <server\_ip>

Specify the IP address of the onvif server, if not specified, the onvif server will listen to all network interfaces.

### <http\_enable>

Indicates whether enable http server, 0-disable, 1-enable.

### <http\_port>

Specify the http server port, providing onvif web service on this port, the default is 8000.

**Note:** On Linux systems, ports below 1024 are reserved by the system and require root privileges to be used.

### <https\_enable>

Indicates whether enable https server, 0-disable, 1-enable.

### <https\_port>

Specify the https server port, providing onvif web service on this port, the default is 8443.

**Note:** On Linux systems, ports below 1024 are reserved by the system and require root privileges to be used.

### <cert\_file>

If HTTPS is enabled (<https\_enable> set to 1), specify the SSL certificate file.

### <key\_file>

If HTTPS is enabled (<https\_enable> set to 1), specify the SSL private key file.

---

**Note:** The certificate file cert.pem and key file key.pem provided by default are self-signed local hosts certificates, only for testing purposes (browsers may pop up untrusted certificate warnings) and cannot be used in formal deployment environments.

**<http\_max\_users>**

Maximum supported HTTP client numbers, if both HTTP and HTTPS are enabled, they can support  $2 * \text{http\_max\_users}$  connections in total.

**<ipv6\_enable>**

Indicates whether IPv6 is enabled, 0-disable, 1-enable.

**Note:** If the device does not specify a server ip in **<server\_ip>** and the **<ipv6\_enable>** is 1, and the device has an IPv6 address, the client can connect to the device through the IPv6 address.

**<need\_auth>**

Indicates whether authentication is required, 0-don't require, 1-require.

**<log\_enable>**

Indicates whether logging is enabled, 0-disable, 1-enable.

**<log\_level>**

The log level:

|       |   |
|-------|---|
| TRACE | 0 |
| DEBUG | 1 |
| INFO  | 2 |
| WARN  | 3 |
| ERROR | 4 |
| FATAL | 5 |

**<information>:** Config the ONVIF device basic information.

**<Manufacturer>**

The manufactor of the device.

---

**<Model>**

The device model.

**<FirmwareVersion>**

The firmware version of the device.

**<SerialNumber>**

The serial number of the device.

**<HardwareId>**

The hardware ID of the device.

**<user>**: Contains a list of the onvif users, it can configure multiple nodes.

**<username>**

Username string.

**<password>**

Password string.

**<userlevel>**

User level string, the following values can be configured:

**Administrator**

**Operator**

**User**

**Anonymous**

**<profile>**: A media profile maps a video and audio source to a video and audio encoder configurations. It can configure multiple nodes.

Currently, a maximum of 8-10 profiles are supported, because too many profiles will result in too large GetProfiles response messages.

**<video\_source>**: If the media profile contains video, the video source configuration.

---

**<width>**

The video source width.

**<height>**

The video source height.

**<video\_encoder>**: If the media profile contains video, the video encoder configuration.

**<width>**

Encoded video width.

**<height>**

Encoded video height.

**<quality>**

Relative value for the video quantizers and the quality of the video.  
A high value within supported quality range means higher quality.

**<session\_timeout>**

The rtsp session timeout for the related video stream.

**<framerate>**

Maximum output framerate in fps.

**<encoding\_interval>**

Interval at which images are encoded and transmitted. (A value of 1 means that every frame is encoded, a value of 2 means that every 2nd frame is encoded ...).

**<bitrate\_limit>**

The maximum output bitrate in kbps.

**<encoding>**

Video codec, either JPEG, MPEG4, H264 or H265.

---

<h264>: Configure H.264 related parameters.

<gov\_length>

Group of Video frames length. Determines typically the interval in which the I-Frames will be coded. An entry of 1 indicates I-Frames are continuously generated. An entry of 2 indicates that every 2nd image is an I-Frame, and 3 only every 3rd frame, etc. The frames in between are coded as P or B Frames.

<h264\_profile>

The H.264 profile, either Baseline, Main, Extended or High.

<h265>: Configure H.265 related parameters.

<gov\_length>

Group of Video frames length. Determines typically the interval in which the I-Frames will be coded. An entry of 1 indicates I-Frames are continuously generated. An entry of 2 indicates that every 2nd image is an I-Frame, and 3 only every 3rd frame, etc. The frames in between are coded as P or B Frames.

<h265\_profile>

The H.265 profile, either Main or Main10.

<mpeg4>: Configure MPEG4 related parameters.

<gov\_length>

Determines the interval in which the I-Frames will be coded. An entry of 1 indicates I-Frames are continuously generated. An entry of 2 indicates that every 2nd image is an I-Frame, and 3 only every 3rd frame, etc. The frames in between are coded as P or B Frames.

<mpeg4\_profile>

The Mpeg4 profile, either simple profile (SP) or advanced simple profile (ASP).

<audio\_source>: If the media profile contains audio, the audio source configuration.

---

**<audio\_encoder>**: If the media profile contains audio, the audio encoder configuration.

**<session\_timeout>**

The rtsp session timeout for the related audio stream.

**<sample\_rate>**

The output sample rate in kHz.

**<bitrate>**

The output bitrate in kbps.

**<encoding>**

Audio codec used for encoding the audio input (either G711, G726 or AAC).

**<stream\_uri append\_params="0">**

The RTSP stream address of the profile, if not specify, the default is:

**rtsp://yourip/test.mp4**

The **append\_params** attribute specifies whether to append audio and video encoding parameters to the end of the rtsp stream. If the stream\_uri attribute does not specify a rtsp stream address, the default rtsp stream address will append audio and video encoding parameters regardless of whether append\_params is 0 or 1. The format of the appended parameters is as follows:

&params=value

The supported params as follows:

t, transmission mode, taking the value of unicast to represent unicast or multicast to represent multicast.

p, transmission protocol, value udp, tcp, rtsp, http.

ve, video encoding, value JPEG, MP4V-ES, H264, H265.

w, video width.

h, video height.

ae, audio encoding, value PCMU, G726, MP4A-LATM (AAC).

sr, audio sample rate.

For example:

rtsp://127.0.0.1/test.mp4&t=unicast&p=udp&ve=H264&w=1280&h=720&ae=PCMU&

---

sr=8000

Indicates UDP unicast mode, video encoding is H264, video resolution is 1280\*720, audio encoding is PCMU, sampling rate is 8K.

#### **<scope>**

Contains a list of URI defining the device scopes.

All ONVIF defined scope URIs have the following format:

onvif://www.onvif.org/<path>

A device may have other scope URIs. These URIs are not restricted to ONVIF defined scopes.

A device shall include at least one fixed entry (defined by the device vendor) of the profile, hardware and name categories respectively in the scopes list. A device may include any other additional scope attributes in the scopes list.

A device might include an arbitrary number of scopes in its scope list. This implies that one unit might for example define several different location scopes. A probe is matched against all scopes in the list.

#### **<event>:** Event Configuration parameters

##### **<renew\_interval>**

Events renew interval.

The onvif client subscribes or creates an event polling point. If the renew or pullmessage request is not called within the renew\_interval interval, the onvif server will delete the subscription or event polling point.

##### **<simulate\_enable>**

Specifies whether to generate simulation event, 0-disable, 1-enable.

---

## Chapter 3 Configuration file

When run `onvif` server for the first time, use the default configuration file `onvif.cfg`, which sets 2 profiles.

When stop `onvif` server, it writes the runtime configurations into the `onvifrun.cfg` file, and the configurations in the `onvifrun.cfg` file will be load at the next time it runs.

If you modify the default configuration file `onvif.cfg`, you should stop the `onvif` server first, then delete the runtime configuration `onvifrun.cfg`, and run `onvif` server again to make the default configuration effective.

If you are familiar with the runtime configuration file `onvifrun.cfg`, you can also modify it.

---

## Chapter 4 Compatibility test

ONVIF SERVER PROFILE S passed the compatibility test version.

Windows versions download from:

<https://www.happytimesoft.com/downloads/happytime-onvif-server-profiles.zip>

Linux versions download from:

<https://www.happytimesoft.com/downloads/happytime-onvif-server-profiles.tar.gz>

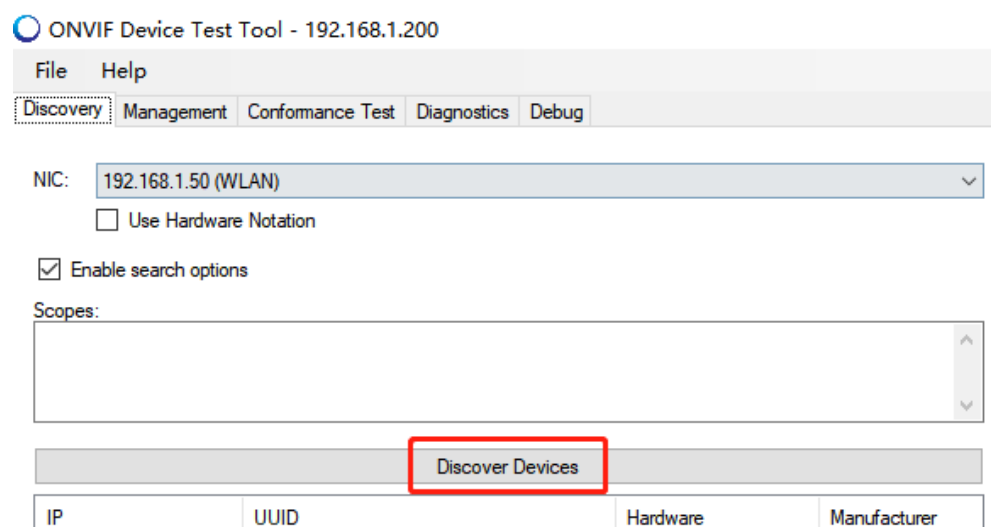
Follow the steps below to perform compatibility testing.

1. Modify the ONVIF SERVER configuration file `onvif.cfg` and specify the `<need_auth>` value to 1.
2. If there is an onvif runtime configuration file, delete the runtime configuration file `onvifrun.cfg`.
3. Run the rtsp server and onvif server.
4. Run the ONVIF Device Test Tool.

**Note:** ONVIF SERVER and test tools should run on different computers. On the same computer, there may be a port conflict as both the onvif server and testing tool require the use of device discovery port 3702.

**Note:** Please disable the firewall on the computer running the testing tool so that the event subscription test cases can pass through.

5. Click “Discover Devices” button, as the following:



6. Switch to “Management” tab, select “Events” tab, then click “Get” button, as the following:

ONVIF Device Test Tool - 192.168.3.100

File Help

Discovery Management **Conformance Test** Diagnostics Debug

Timeouts

Message Timeout (ms): 30000

Reboot Timeout (ms): 30000

Time Between Tests (ms): 0

Operation Delay (ms): 1000

Time Between Requests (ms): 0

Environment

DNS IPv4: 10.1.1.1 IPv6: 2001:1:1:1:

NTP IPv4: 10.1.1.1 IPv6: 2001:1:1:1:

GW IPv4: 10.1.1.1 IPv6: 2001:1:1:1:

Multicast Address Range

☐ Use multicast address range

IPv4 Min: 239.0.0.0 Max: 239.255.255.255

IPv6 Min: ff15:0:0:0:0:0:0:0 Max: ff15:0:0:0:0:0:0:0

Miscellaneous

Security PTZ Events I/O Recording QuickInstall Credential Identifier Values Access Control Settings Operator Actions

First event topic: Ins1.Device/Trigger/DigitalInput

Topic namespaces: Ins1="http://www.onvif.org/ver10/topics"

Second event topic: Ins1.Device/Trigger/DigitalInput

Second topic namespaces: Ins1="http://www.onvif.org/ver10/topics"

☒ Show only property events

Subscription Timeout (s): 60

Get

7. select “Recording” tab, then click “Get” button, as the following:

ONVIF Device Test Tool - 192.168.3.100

File Help

Discovery Management Conformance Test **Diagnostics** Debug

Timeouts

Message Timeout (ms): 30000

Reboot Timeout (ms): 30000

Time Between Tests (ms): 0

Operation Delay (ms): 1000

Time Between Requests (ms): 0

Environment

DNS IPv4: 10.1.1.1 IPv6: 2001:1:1:1:

NTP IPv4: 10.1.1.1 IPv6: 2001:1:1:1:

GW IPv4: 10.1.1.1 IPv6: 2001:1:1:1:

Multicast Address Range

☐ Use multicast address range

IPv4 Min: 239.0.0.0 Max: 239.255.255.255

IPv6 Min: ff15:0:0:0:0:0:0:0 Max: ff15:0:0:0:0:0:0:0

Miscellaneous

Security PTZ Events I/O **Recording** QuickInstall Credential Identifier Values Access Control Settings Operator Actions

Recording for tests: RECORDING\_405

Timeout for getting results (s): 10

Metadata Filter:

Retention Time (duration): P1D

Get

Validate

8. Switch to “Conformance Test” tab, click “Start Conformance Test” button:

ONVIF Device Test Tool - 192.168.1.200

File Help

Discovery Management **Conformance Test** Diagnostics Debug

Responsible Member

\* Member Name: Happytimesoft

\* Member Address: shenzhen,china Clear

Device Under Test Information

\* Product Name: Happytime onvif server

Brand: Happytimesoft

Model: IPCamera

Other Information:

\* Product Type:

☐ Access Controller

☐ Access Controller Gateway

☐ Access Control Management System

☐ Decoder

☐ Encoder

Product Type (other): << Add

Clear

Technical Support Information Test Execution Information Authorized Representative Signature Settings

\* Technical support website URL: http://www.happytimesoft.com

Technical support e-mail:

Technical support phone:

\* General international support mailing address: support@happytimesoft.com

Regional support contact address:

Clear

Conformance

▶▶ Start Conformance Test

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## Chapter 5 ONVIF features

The onvif server supports the onvif features listed in the following table:

| Feature        |                          |                                         |
|----------------|--------------------------|-----------------------------------------|
| Security       | WS-Username Token        |                                         |
|                | Digest                   |                                         |
| Discovery      | BYE Message              |                                         |
|                | Types                    | tds:Device                              |
|                |                          | dn:Network Video Transmitter            |
| Device Service | Capabilities             | GetCapabilities                         |
|                |                          | GetService                              |
|                | Network                  | Zero Configuration                      |
|                |                          | NTP                                     |
|                |                          | Dynamic DNS                             |
|                |                          | IP Filter                               |
|                |                          | HTTPS                                   |
|                | System                   | System Logging                          |
|                |                          | HTTP System Logging                     |
|                |                          | HTTP Firmware Upgrade                   |
|                |                          | HTTP Support Information                |
|                |                          | HTTP System Backup                      |
|                | Security                 | Default Access Policy                   |
|                |                          | Maximum Users                           |
|                |                          | Remote User Handling                    |
|                |                          | Maximum Username Length                 |
|                |                          | Maximum Password Length                 |
|                | I/O                      | Relay outputs                           |
| Event Service  | WS Basic Notification    |                                         |
|                | Message Content Filter   | ONVIF Message Content Filter<br>Dialect |
|                | Get Service Capabilities | MaxPullPoints capability                |
|                | Pull-Point Notification  |                                         |
| Media Service  | Video                    | JPEG                                    |
|                |                          | H.264                                   |
|                |                          | MPEG4                                   |
|                | Audio                    | G.711                                   |
|                |                          | G.726                                   |

|                |                      |                            |
|----------------|----------------------|----------------------------|
|                |                      | AAC                        |
|                | Audio Output         | G.711                      |
|                |                      | AAC                        |
|                | Real-time Streaming  | RTP/UDP                    |
|                |                      | RTP/RTSP/HTTP              |
|                |                      | RTP/RTSP/TCP               |
|                |                      | RTP-Multicast/UDP          |
|                | Snapshot URI         |                            |
| Media2 Service | Video                | H.265                      |
|                |                      | H.264                      |
|                | Audio                | G.711                      |
|                |                      | AAC                        |
|                | Audio outputs        | G.711                      |
|                |                      | AAC                        |
|                | Real-time Streaming  | RTP/UDP                    |
|                |                      | RTP/RTSP/HTTP              |
|                |                      | RTP/RTSP/TCP               |
|                |                      | RTP-Multicast/UDP          |
|                | RTSP WebSocket       |                            |
|                | Snapshot URI         |                            |
|                | Video Source Mode    |                            |
|                | OSD                  |                            |
|                | Analytics            |                            |
|                | Metadata             |                            |
|                | Media2 Events        | Media/ProfileChanged       |
|                |                      | Media/ConfigurationChanged |
| PTZ Service    | Absolute move        | Pan/Tilt movement          |
|                |                      | Zoom movement              |
|                | Relative move        | Pan/Tilt movement          |
|                |                      | Zoom movement              |
|                | Continuous move      | Pan/Tilt movement          |
|                |                      | Zoom movement              |
|                | Presets              |                            |
|                | Home position        | Configuration              |
|                | Auxiliary operations |                            |
|                | Speed                | Speed for Pan/Tilt         |
|                |                      | Speed for Zoom             |

|                           |                                     |                             |
|---------------------------|-------------------------------------|-----------------------------|
|                           | Move Status                         |                             |
|                           | Status Position                     |                             |
|                           | Get Compatible Configurations       |                             |
| Device IO Service         | Relay outputs                       | Bistable Mode               |
|                           |                                     | MonoStable Mode             |
|                           | Digital Inputs                      | Digital Input Options       |
| Imaging Service           | IrCutfilter Configuration           |                             |
|                           | Tampering Events                    | Image Too Blurry            |
|                           |                                     | Image Too Dark              |
|                           |                                     | Image Too Bright            |
|                           |                                     | Global Scene Change         |
|                           | Motion Alarm                        |                             |
| Analytics Service         | Rule Engine                         | Rule Options                |
|                           |                                     | Motion Region Detector Rule |
|                           | Analytics Modules                   | Analytics Module Options    |
| Recording Control Service | Dynamic Recordings                  |                             |
|                           | Dynamic Tracks                      |                             |
|                           | Audio Recording                     |                             |
|                           | Recording Options                   |                             |
|                           | tns1:RecordingCofig/DeleteTrackData |                             |
|                           | Metadata Recording                  |                             |
|                           | Encoding                            | JPEG                        |
|                           |                                     | H264                        |
|                           |                                     | MPEG4                       |
| Recording Search Service  | Metadata Search                     |                             |
|                           | PTZ Position Search                 |                             |
| Door Control Service      | Door Entity                         | Access Door                 |
|                           |                                     | Lock Door                   |
|                           |                                     | Double Lock Door            |
|                           |                                     | Block Door                  |
|                           |                                     | Lock Down Door              |
|                           |                                     | Lock Open Door              |
|                           |                                     | Door Monitor                |
|                           |                                     | Double Lock Monitor         |
|                           |                                     | Alarm                       |

|                        |                                    |                             |
|------------------------|------------------------------------|-----------------------------|
|                        |                                    | Tamper                      |
|                        |                                    | Fault                       |
|                        | Door Control Events                |                             |
|                        | Door Management                    |                             |
|                        | Client Supplied Token              |                             |
| Access Control Service | Area Entity                        |                             |
|                        | Access Point Entity                | Enable/Disable Access Point |
|                        |                                    | Duress                      |
|                        |                                    | Access Taken                |
|                        |                                    | Anonymous Access            |
|                        | Access Point Management            |                             |
|                        | Area Management                    |                             |
|                        | Access Control Events              |                             |
| Replay Service         | RTP/RTSP/TCP                       |                             |
| Receiver Service       |                                    |                             |
| Credential Service     | Credential Validity                |                             |
|                        | Credential Access Profile Validity |                             |
|                        | pt:Card                            |                             |
|                        | pt:PIN                             |                             |
|                        | pt:Fingerprint                     |                             |
|                        | Reset Antipassback Violation       |                             |
|                        | Client Supplied Token              |                             |
|                        | Whitelist                          |                             |
|                        | Blacklist                          |                             |
|                        | Validity Supports Time Value       |                             |
| Access Rules Service   | Multiple Schedules Access Point    |                             |
|                        | Client Supplied Token              |                             |
| Schedule Service       |                                    |                             |
| Thermal Service        |                                    |                             |

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## Chapter 6 ONVIF Version

The onvif server implements the following ONVIF service:

| ONVIF Service     | Prefix | Url                                                                                                             | version |
|-------------------|--------|-----------------------------------------------------------------------------------------------------------------|---------|
| device            | tds    | <a href="http://www.onvif.org/ver10/device/wsdl">http://www.onvif.org/ver10/device/wsdl</a>                     | 23.06   |
| event             | tev    | <a href="http://www.onvif.org/ver10/events/wsdl">http://www.onvif.org/ver10/events/wsdl</a>                     | 22.06   |
| media             | trt    | <a href="http://www.onvif.org/ver10/media/wsdl">http://www.onvif.org/ver10/media/wsdl</a>                       | 21.12   |
| media 2           | tr2    | <a href="http://www.onvif.org/ver20/media/wsdl">http://www.onvif.org/ver20/media/wsdl</a>                       | 23.06   |
| ptz               | tptz   | <a href="http://www.onvif.org/ver20/ptz/wsdl">http://www.onvif.org/ver20/ptz/wsdl</a>                           | 22.12   |
| image             | timg   | <a href="http://www.onvif.org/ver20/imaging/wsdl">http://www.onvif.org/ver20/imaging/wsdl</a>                   | 22.06   |
| analytics         | tan    | <a href="http://www.onvif.org/ver20/analytics/wsdl">http://www.onvif.org/ver20/analytics/wsdl</a>               | 22.06   |
| recording control | trc    | <a href="http://www.onvif.org/ver10/recording/wsdl">http://www.onvif.org/ver10/recording/wsdl</a>               | 23.06   |
| search            | tse    | <a href="http://www.onvif.org/ver10/search/wsdl">http://www.onvif.org/ver10/search/wsdl</a>                     | 22.06   |
| replay            | trp    | <a href="http://www.onvif.org/ver10/replay/wsdl">http://www.onvif.org/ver10/replay/wsdl</a>                     | 21.12   |
| access control    | tac    | <a href="http://www.onvif.org/ver10/accesscontrol/wsdl">http://www.onvif.org/ver10/accesscontrol/wsdl</a>       | 21.06   |
| door control      | tdc    | <a href="http://www.onvif.org/ver10/doorcontrol/wsdl">http://www.onvif.org/ver10/doorcontrol/wsdl</a>           | 21.06   |
| device IO         | tmd    | <a href="http://www.onvif.org/ver10/deviceIO/wsdl">http://www.onvif.org/ver10/deviceIO/wsdl</a>                 | 22.06   |
| thermal           | tth    | <a href="http://www.onvif.org/ver10/thermal/wsdl">http://www.onvif.org/ver10/thermal/wsdl</a>                   | 22.06   |
| credential        | tcr    | <a href="http://www.onvif.org/ver10/credential/wsdl">http://www.onvif.org/ver10/credential/wsdl</a>             | 21.06   |
| access rules      | tar    | <a href="http://www.onvif.org/ver10/accessrules/wsdl">http://www.onvif.org/ver10/accessrules/wsdl</a>           | 19.06   |
| schedule          | tsc    | <a href="http://www.onvif.org/ver10/schedule/wsdl">http://www.onvif.org/ver10/schedule/wsdl</a>                 | 18.12   |
| receiver          | trv    | <a href="http://www.onvif.org/ver10/receiver/wsdl">http://www.onvif.org/ver10/receiver/wsdl</a>                 | 21.12   |
| provisioning      | tpv    | <a href="http://www.onvif.org/ver10/provisioning/wsdl">http://www.onvif.org/ver10/provisioning/wsdl</a>         | 18.12   |
| security          | tas    | <a href="http://www.onvif.org/ver10/advancedsecurity/wsdl">http://www.onvif.org/ver10/advancedsecurity/wsdl</a> | 24.12   |

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## Chapter 7 Supports multiple channels

The onvif server supports multi-channel. Each <profile> tag represents a channel in the configuration file.

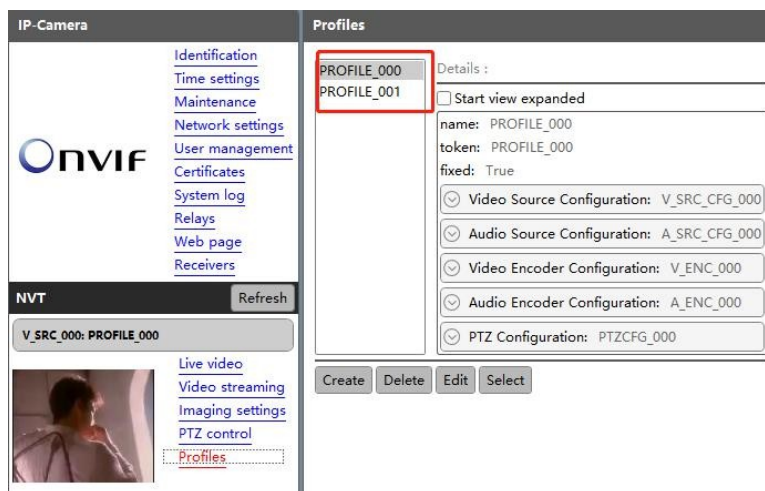
The default configuration file supports 2 channels, you can add <profile> tag to support more channels.

Note: If <video\_source>.width and <video\_source>.height of multiple <profile> tags are the same, it is considered that they are using the same video source, example:

```
<profile>
  <video_source>
    <width>1280</width>
    <height>720</height>
  </video_source>
  ....
</profile>
```

```
<profile>
  <video_source>
    <width>1280</width>
    <height>720</height>
  </video_source>
  ....
</profile>
```

The onvif device manager will show the profiles as the following:



If `<video_source>.width` and `<video_source>.height` of multiple `<profile>` tags are not the same, it is considered that they are using different video sources, example:

```
<profile>
  <video_source>
    <width>1280</width>
    <height>720</height>
  </video_source>
  ....
</profile>
```

```
<profile>
  <video_source>
    <width>640</width>
    <height>480</height>
  </video_source>
  ....
</profile>
```

The onvif device manager will show the profiles as the following:



[Identification](#)[Time settings](#)[Maintenance](#)[Network settings](#)[User management](#)[Certificates](#)[System log](#)[Relays](#)[Web page](#)[Receivers](#)

NVT

Refresh

V\_SRC\_000: PROFILE\_000



[Live video](#)[Video streaming](#)[Imaging settings](#)[PTZ control](#)[Profiles](#)

V\_SRC\_001: PROFILE\_001



[Live video](#)[Video streaming](#)[Imaging settings](#)[PTZ control](#)[Profiles](#)

Profiles

PROFILE\_000

PROFILE\_001

---

## Chapter 8 Modify RTSP stream address

If the value of `<stream_uri>` in the `<profile>` tag in the onvif server configuration file is not modified, the RTSP stream address provided by the onvif server by default is `rtsp://ip/test.mp4`, you can modify the `<stream_uri>` in `<profile>` tag to specify the rtsp stream address provided by the onvif server. such as:

```
<profile>
...
<stream_uri append_params="0">rtsp://192.168.3.27/live</stream_uri>
</profile>
```

---

## Chapter 9 Run Onvif Server

Windows platform:

Run runme.bat, it will run rtsp server and onvif server.

Linux platform:

Run runme.sh, it will run rtsp server and onvif server.